EX. NO: 01	Database Design Using Conceptual Modelling (ER Diagram)

ER - Diagram [Entity Relationship Diagram]

ER Model describes data in terms of:

* Entities and entity sets
  objects
* Relationships and relationship sets
  connection between objects
* Attributes
  properties that characterize or describe entities or relationships

ENTITY SETS:

Entity type or set

* Collection (or set) of similar entities that have the same attributes.

Categories of Attributes:

1. Simple (atomic) vs. composite attributes.
2. Single - valued vs. Multivalued attributes.
3. Stored vs. derived attributes
4. Key or unique attributes

→ Attribute values constrained to be distinct for individual entities in entity set

Database life cycle - Logical design:

1) Top down approach (Entity - Relationship (ER) model)

- This approach is used when application requirements are clear.
- Represents the application requirements in a pictorial form.
- The real word objects and their corresponding attribute are identified from the requirements - hence it is top down.

- This model helps in
  - → analysis and design
  - → re-validating the requirements.

2) Bottom up approach (Normalization)

* This approach is used when application requirements are not very clear

* First define the required data items and then group the related data items

* Further refinement may be carried out depending on the application need.

Case Study: Designing a Database for Easy shop - Retail Application

1. One supplier can provide many quotations
2. For every item, there will be a separate quotation.
3. Many suppliers can quote for the same time
4. Many items can be outward to the several retail outlets
5. One employee works for only one retail outlet.
6. One retail outlet can have many employees.
7. Many customers can purchase items from several retail outlets.
8. One employee can nominate many dependents.

Steps in E-R Diagram:

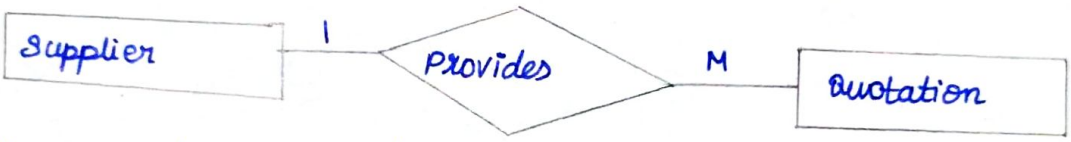
- Identify the entities
- Find the relationships and cardinality
- Identify the key attributes
- Identify other relevant attributes
- Draw the ER diagram.

Step 1 : Identify the entities.

- Supplier
- Quotation
- Item
- Retail Outlet
- Employee
- Customer
- Dependent

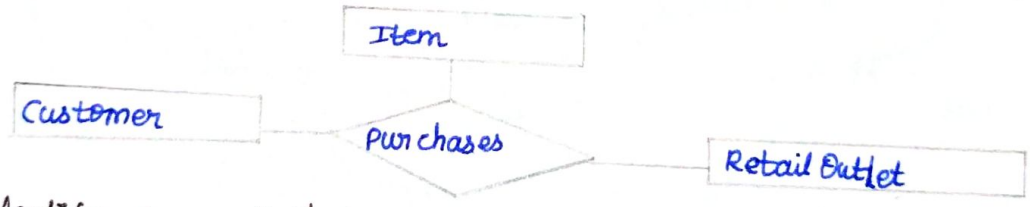
Step 2 : Find relationship and cardinality.

One supplier can provide many quotations.



Many items can be outward to several retail outlets.

Many customers can purchase items from several retail outlets



Step 3 : Identify key attributes

Supplier - supplierid

Quotation - quotationid

Item - itemcode

RetailOutlet - retailoutletid

Employee - empid

Customer - customerid.

Step 4 : Identify other relevant attributes.

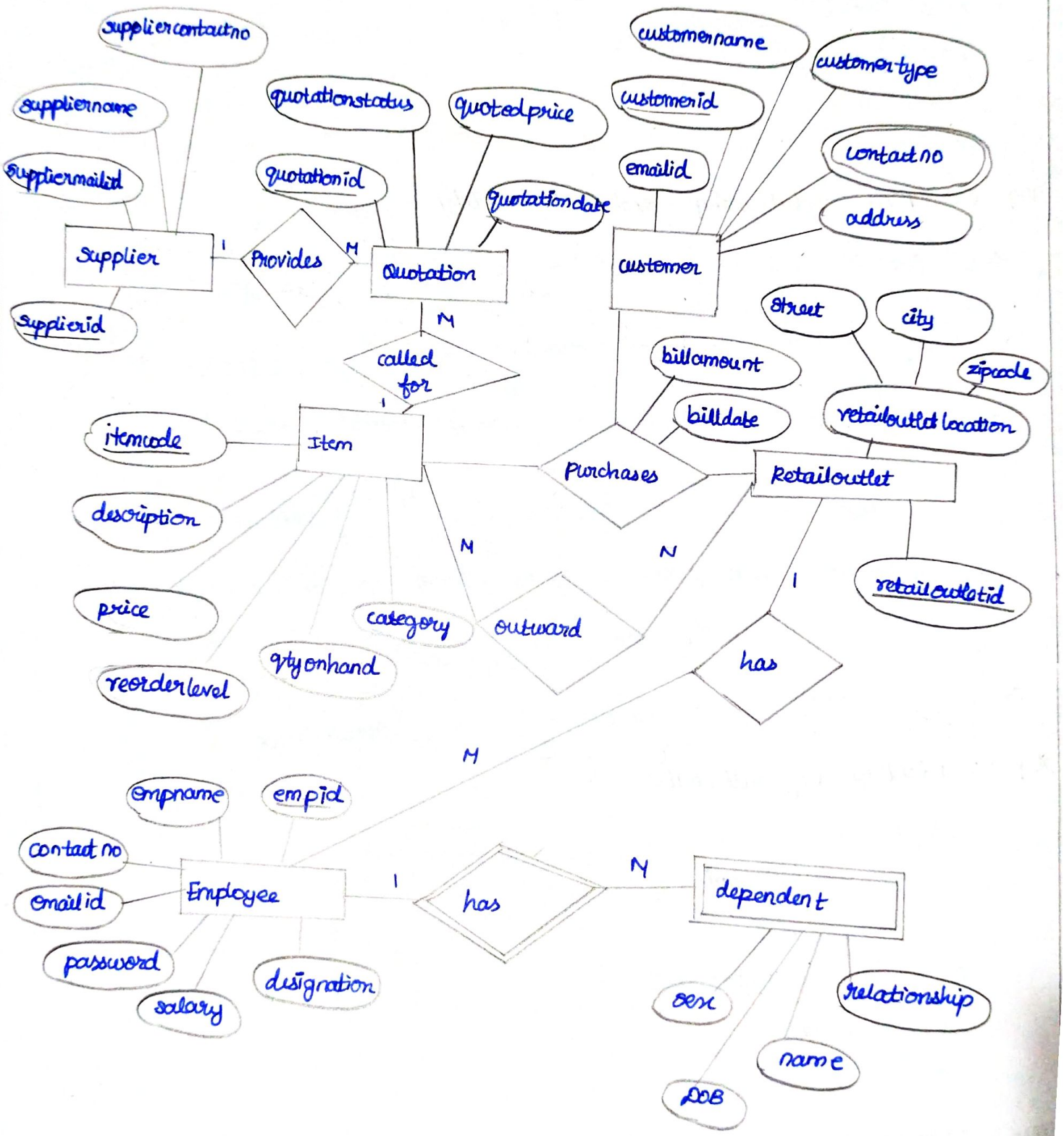
Supplier : suppliername

Quotation : quotedprice, quotationdate, quotationstatus.

Item : Itemcode, description, price, category, qtyonhand, reorderlevel

RetailOutlet : retailoutletlocation

Employee: empname, designation, emailId, contactno, salary, password.

customer: customertype, customername, email id, contactno, address

STEP 5: Draw the ER Diagram.